Matthew's Star of Bethlehem is inconsistent with natural objects: evidence from orbital mechanics and corpus linguistics

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Abstract

Matthew 2:9 describes the Star of Bethlehem as going before travelers to Bethlehem and then standing over the place where the child was. This study asks whether any natural object could produce that description. A survey of 529 million heliocentric configurations, spanning all orbit types and every proposed epoch (12–1 BCE), finds none that guides along the road corridor and then stops. Monte Carlo refinement, a geocentric survey and close-flyby campaigns over $\mathbf{10^9}$ sampled orbits, including loosened criteria and two-leg itineraries, give the same null result, for kinematic reasons. Independently, Matthew's four motion terms (προάγω, ἵστημι, ἐπάνω, ἔρχομαι) are absent in his sense from 24 Greek astronomical texts ($\approx$**619,000** words; Fisher combined $\boldsymbol{p \approx 4.5 \times 10^{-5}}$) but characterize guiding-star narratives (Bayes factor $\boldsymbol{\approx 3 \times 10^6}$). The Star as Matthew describes it is inconsistent with a natural celestial object and is better explained as legendary narrative.

Keywords: Star of Bethlehem, orbital mechanics, corpus linguistics, Matthew 2:9, archaeoastronomy, ancient Greek astronomy

Introduction

Testing whether a historical narrative records a real celestial event is ordinarily a search problem: which known sky object best fits the account? Here it is shown that such questions can instead be tested by *demonstration*: by surveying orbital parameter space exhaustively against explicit criteria drawn from the text and asking whether

any configuration satisfies them. I apply this paradigm to the most studied ancient astronomical claim in the Western record and show that two independent lines of evidence, orbital mechanics and corpus linguistics, converge on the same verdict.

The Star of Bethlehem debate was placed on a renewed scientific footing when Hughes proposed in these pages that the triple Jupiter–Saturn conjunction of 7 BCE was the most probable candidate[1, 2], a paper that has accumulated more than 400 citations and sustained the scientific conversation for five decades. Other proposals include Halley's Comet[3], a nova[4, 5], the retrograde motion of Jupiter in Aries[6], and a 5 BCE comet whose possible orbital elements have most recently been reconstructed from East Asian records[7, 8]. The present survey explicitly encompasses Matney's proposed orbital parameters (Monte Carlo Sweep 2; see Methods), and this specific candidate also fails the criteria. Systematic reviews show no single candidate satisfies all criteria simultaneously[9, 10], and the majority position in New Testament criticism holds that Matthew's infancy narrative is legendary rather than historical[11–13], though active discussion remains within biblical studies (cf. [14–16]). What has been lacking is not a better candidate but a general test of whether any orbital solution of any class can satisfy the text's kinematic requirements, and a quantitative test of whether Matthew's vocabulary belongs to the astronomical register at all.

The critical constraints come from Matthew 2:9: ὁ ἀστὴρ οὗ εἶδον ἐν τῇ ἀνατολῇ προῆγεν αὐτούς, ἕως ἐλθὼν ἐστάθη ἐπάνω οὗ ἦν τὸ παιδίον ("the Star which they had seen at its rising was going before them, until, having arrived, it stood over where the child was"). The verb προῆγεν (imperfect active of προάγω) denotes sustained forward motion ahead of the travelers throughout their journey. The form ἐστάθη (aorist passive of ἵστημι), bounded by the temporal particle ἕως ("until"), denotes a completed, punctual coming-to-rest, a change of state, not a prior condition[9]. The phrase ἐπάνω οὗ ("over where") further specifies that the stopping occurred directly above a particular dwelling[9, 17, 18]. The two verbs define two physically distinct phases of apparent motion: a *guidance phase* in which the Star moves ahead of travelers walking from Jerusalem to Bethlehem, and a *stopping phase* in which it becomes motionless above a specific house.

Three converging lines of evidence are presented in the order that builds the argument most naturally. First, Matthew's own usage of the four motion words across his other 118 Gospel occurrences, zero in any astronomical sense, yields an internal Bayes factor of $\approx 88{,}000$ against the astronomical reading. Second, orbital mechanics addresses the loose-language escape: the incompatibility is geometric. Objects in the road corridor appear quasi-stationary (satisfying the stopping criterion but not guidance), while objects with sufficient southward motion lie outside it; no configuration tested satisfies both. Third, external corpus-linguistic analysis of 24 Greek astronomical texts ($\approx 619{,}000$ words) places Matthew's vocabulary in the legendary guiding-star tradition. The linguistic datasets are disjoint, and their joint Bayes factor remains decisive across the full range of dependence assumptions tested, from $\approx 6 \times 10^3$ under the most conservative to $\approx 2.6 \times 10^{11}$ under strict independence.

Methods

Reference epoch, site, and coordinate system

All calculations used the International Celestial Reference System (ICRS, J2000.0) as the inertial frame. The reference observation window is Julian Day 1 719 755.9 (TDB), corresponding to ≈ 08:00 local solar time at Bethlehem (31.70° N, 35.20° E, 765 m) on a date in 5 BCE. The UT1–UTC offset (ΔT = 10,572 s) was derived from the long-term IERS extrapolation of Morrison & Stephenson[19]. Jerusalem Old City center (31°46′ N, 35°14′ E) to the Church of the Nativity, Bethlehem (31°42′ N, 35°12′ E) gives a straight-line road bearing of 203°; the ridge road sweeps the corridor $[190°, 210°]$ throughout the journey.

Transformation from ICRS to local horizontal (azimuth–altitude) coordinates used pre-computed East–North–Up (ENU) rotation matrices constructed via `astropy`[20] `AltAz`→`SkyCoord.icrs`, incorporating Earth's precession, nutation, and Greenwich Apparent Sidereal Time at each time step. Applying J2000.0 hour-angle formulae without precession correction at 5 BCE produces a 25.7° systematic azimuth error; the ENU approach eliminates this. The matrices were validated by recovering the azimuth of Matney's proposed 5 BCE comet[8] (193.7°) to within 0.01° at the reference epoch.

Orbit propagation

Heliocentric positions were computed from osculating orbital elements for each orbit type. For parabolic orbits ($e = 1$), Barker's equation gives the true anomaly directly. For elliptic ($0 \le e < 1$) and hyperbolic ($e > 1$) orbits, the mean anomaly $M = n(t - T)$ (or its hyperbolic analogue) was iterated to Newton–Raphson convergence at tolerance 10^{-12} rad, then converted to true anomaly. For the AU-scale surveys, Earth positions came from the ephemeris built into `astropy` (ERFA `epv00`); at 5 BCE its error displaces sightlines by $\le 0.024°$ for objects beyond 0.5 AU and $\le 0.12°$ at 0.1 AU, negligible against the 20° corridor. Every analysis of objects within 0.05 AU (the close-flyby campaigns) instead takes Earth's position from VSOP87, since no modern JPL ephemeris extends to 5 BCE and the error matters at the degree level at lunar distances (Supplementary Note 3). Geocentric configurations used the same solver with $GM_\oplus = 3.986 \times 10^5\,\mathrm{km^3\,s^{-2}}$, with the observer's GCRS position from `astropy` at each time step. Heliocentric positions were vectorised over Ω and ω using pre-computed perifocal unit vectors with `NumPy`[21] `einsum`, enabling evaluation of $72 \times 72 = 5{,}184$ orientations per loop iteration at 5° grid resolution.

Guidance and stopping criteria

Three criteria operationalise Matthew's text. (1) *Visibility:* object altitude > 5° throughout the guidance window (08:00–10:00 local time, nine 15-min steps). (2) *Azimuth:* object azimuth remains within $[190°, 210°] \pm 5°$ at every step, the road corridor from Jerusalem to Bethlehem. (3) *Motion:* guidance phase (προῆγεν, ongoing) requires minimum apparent angular velocity $> \omega_{\mathrm{thresh}} = 2°\,\mathrm{h^{-1}}$; stopping phase (ἐστάθη, completed) requires maximum apparent angular velocity $< \omega_{\mathrm{thresh}}$ during 10:15–12:00 (eight steps). In this grid survey the angular velocities are measured

against the stellar background, from the object's geocentric direction in the ICRS. The azimuth criterion is necessarily evaluated in the local horizontal frame, in which every such body also shares the diurnal rotation of $\approx 15^\circ\,\mathrm{h}^{-1}$ westward. The literal, ground-frame reading of "going before" and "stood" is treated separately, by the kinematic argument and the close-flyby campaigns (Table 1).

Because the grid survey measures motion against the stars, the Moon's motion against the stars is the appropriate comparison for its threshold. The threshold $\omega_{\mathrm{thresh}} = 2^\circ\,\mathrm{h}^{-1}$ is four times the lunar rate ($\approx 0.5^\circ\,\mathrm{h}^{-1}$ against background stars) and is detectable by naked eye within minutes; it is deliberately generous, as any stricter value only strengthens the null result. Sensitivity analysis across two orders of magnitude (0.25–$10^\circ\,\mathrm{h}^{-1}$) confirms zero configurations satisfy all three criteria at any tested threshold (Supplementary Table 1). The lower limit ($0.25^\circ\,\mathrm{h}^{-1}$) corresponds to motion of $\approx 5'$ over 20 min, at the practical limit of unaided naked-eye resolution, ensuring the analysis spans every physically detectable level of motion. The upper end of the sweep is independently bounded by a reductio ad absurdum: the Moon moves at $\approx 0.5^\circ\,\mathrm{h}^{-1}$ against the background stars, yet no ancient source describes the Moon as standing still. Any stopping threshold set *above* the lunar rate would therefore classify the Moon itself as stationary, plainly inadmissible. The sensitivity sweep thus brackets the physically meaningful range at both ends: below by the limit of naked-eye resolution, above by the lunar-rate argument. The binding constraint is the azimuth criterion: no Keplerian orbit keeps the object within the road corridor throughout the 2-hour guidance window; the motion criterion provides additional independent exclusion at lower thresholds. Because Ω and ω each span all 360°, the orbital parameter sweep covers all possible sky positions for a body on the reference date, making the null result independent of the specific time of night at which the guidance phase is observed.

The analyses in this paper use different time windows, frames and thresholds, because each addresses a different regime or reading. They are collected in Table 1, so that each number in the text can be traced to the analysis it belongs to. In particular, the 2-hour guidance window belongs to the grid survey; the 1.0-hour guidance requirement belongs to the first close-flyby survey, and the later campaign reduces it to 15 minutes; and the $0.20^\circ\,\mathrm{h}^{-1}$ stopping threshold belongs to the geocentric survey, which works in the ground frame.

Survey design and Monte Carlo refinement

The exhaustive heliocentric survey spanned perihelion distance $q \in [0.02, 1.00]$ AU (18 values), eccentricity e from 0 to 5 (15 values covering elliptic, parabolic, and hyperbolic orbits), inclination i from 0° to 180° in 5° steps plus additional fine values near 0° to 4° (42 total), perihelion epoch offset $\Delta T \in [-60, +60]$ days (9 values), and Ω, ω swept at 5° resolution (72 values each). The outer parameter combinations ($18\times15\times42\times9 = 102{,}060$) multiplied by $72\times72 = 5{,}184$ orientations yield 529,079,040 total configurations, of which 202,372,076 heliocentric configurations were visible at the reference epoch. Zero satisfied all three criteria simultaneously (Supplementary Table 20). A separate geocentric survey spanning LEO to beyond lunar distance

Table 1 Criteria and parameters used in each analysis. "Stars": angular velocity measured against the stellar background (geocentric ICRS direction). "Ground": apparent motion in the local horizontal frame, the literal reading of Matthew 2:9. All azimuth corridors are in the local frame.

Analysis	Frame	Window	Guidance	Stopping	Result
Kinematic argument	Ground, converted to ICRS	08:00–10:00; 10:15–12:00	Rising $\geq 2^\circ\,\mathrm{h}^{-1}$ along az 200°	Fixed in local sky	Requires $V_\perp \gtrsim 10^3\,\mathrm{km\,s}^{-1}$ for $d \gtrsim 0.1\,\mathrm{AU}$
Heliocentric grid, 5.29×10^8	Stars	Daylight; 2 h guide, 1.75 h stop	Corridor $[190^\circ, 210^\circ] \pm 5^\circ$ throughout; $\omega > 2^\circ\,\mathrm{h}^{-1}$ (swept 0.25–10)	$\omega < 2^\circ\,\mathrm{h}^{-1}$ (swept 0.25–10)	0
Monte Carlo refinement, 2.25×10^6	Stars	As grid	As grid	As grid	0
Seven epochs, 12–1 BCE	Stars	As grid	As grid	As grid	0 of 79,788 visible
Geocentric, 28,800	Ground	Night	In corridor; $1 \leq \omega \leq 5^\circ\,\mathrm{h}^{-1}$	$\omega < 0.50$ (strict 0.20)° h^{-1} within 4 h	0 at strict threshold
Close flybys, 10^7 orbits (66 flybys)	Ground	Night (Sun $< -6^\circ$)	≥ 1.0 h in corridor; azimuth drift $< 1.5^\circ\,\mathrm{h}^{-1}$; altitude rate $> 0.3^\circ\,\mathrm{h}^{-1}$	$\omega < 0.30^\circ\,\mathrm{h}^{-1}$ within 4 h	0
Close flybys, 10^9 orbits (6,116 flybys)	Ground	Night	0.25–2 h; corridor 20–60° wide; drift tolerance 1.5–30° h^{-1}	Displacement over 1 h $\leq$ 0.3–5°, within 2–8 h, at the guidance bearing	0 of 525 cells; tightest 6.94°
Two-leg itinerary, same 6,116	Ground	Night	Road leg 0.5–1.5 h, free turn $\leq$ 1 h, approach leg 0.25–0.5 h toward 90–180° or 210–270°	As above, within 8 h	No eastward itinerary; westward: tightest 9.22°

(51,200 configurations; 28,800 with perigee altitude > 100 km) likewise returned zero solutions.

The 5° grid spacing leaves gaps of up to ≈3.5° between neighboring points. To confirm the null result is not a grid artefact, 2.25 million configurations were sampled uniformly in continuous parameter space in two Monte Carlo sweeps: around the best-fit grid orbit (Sweep 1: $e \in [0.7, 2.5]$, $q \in [0.005, 0.15]$ AU, $i \in [0^\circ, 40^\circ]$, 1.5 million configurations) and around Matney's proposed 5 BCE comet parameters (Sweep 2: 750,000 configurations bracketing q, e, i, Ω, ω with generous margins). Of 895,822 visible configurations across both sweeps, zero satisfied all three criteria. The best guidance-phase angular velocity among azimuth-corridor-passing configurations was $0.006^\circ\,\mathrm{h}^{-1}$, a factor of 330 below the $2^\circ\,\mathrm{h}^{-1}$ threshold, confirming that the azimuth–motion trade-off is structural, not an artefact of resolution.

The analysis was repeated at all historically proposed Star of Bethlehem dates (Halley's Comet, 12 BCE; Jupiter–Saturn conjunction, 7 BCE; the 5 BCE reference epoch and surrounding months; 1 BCE nova hypothesis), recomputing Earth positions and ENU matrices at each epoch. Across 79,788 visible configurations at seven epochs, zero satisfied all three criteria (script `sob_epoch_robustness.py`; Supplementary Note 21).

Corpus construction and feature classification

The comparative corpus of 24 ancient Greek astronomical and astrological texts (≈619,000 words, 4th century BCE through 4th century CE) was assembled from the Thesaurus Linguae Graecae (TLG) electronic database[22]. Three genre classes were distinguished. *Mathematical astronomy* (Ptolemy *Almagest* and *Planetary Hypotheses*; Hipparchus; Geminus; Autolycus; Theodosius; Aristarchus; Cleomedes; Hypsicles) covers geometric and predictive description of celestial motions. *Astrological application* (Ptolemy *Tetrabiblos*; Vettius Valens; Hephaestion; Dorotheus; Pseudo-Manetho; Paul of Alexandria) covers celestial-to-human-destiny inference; this class is especially important because it is the genre most likely to employ relational vocabulary of stars guiding or influencing persons: its null result for all four features closes that alternative explanation. *Didactic and popularizing works* (Aratus; Eratosthenes; Achilles Tatius; Theon of Smyrna; Aristotle *Meteorologica*; Theophrastus *De signis*) complete the coverage. The two meteorological works are included because ancient writers treated comets, meteors and weather signs together with celestial phenomena, so the vocabulary of moving lights in the sky is sampled in both genres. Full corpus inventory is given in Supplementary Table 7.

Four lexical features were identified directly from Matthew 2:9, one for each word Matthew uses to describe the Star's motion, and tested against the corpus. Feature A (προάγω): the Star "goes before" the Magi. The corpus contains 11 occurrences; all describe positional or temporal antecedence between celestial bodies (one planet *preceding* another in longitude) and none describe a celestial body actively guiding human travelers toward a destination. Feature B (ἵστημι): the Star "stands". The corpus contains 121 occurrences across sixteen authors. The verb is the ordinary vocabulary of the planetary station (στηριγμός, the moment of zero longitudinal velocity in the geocentric model), and Geminus, Theon of Smyrna and Paul of Alexandria all use it in that sense. What no occurrence supplies is the complement: in every case what a body stands in, on or above is a zodiacal sign, a degree, an aspect, a chart angle or a fixed star, never a terrestrial location. Feature C (ἐπάνω): the Star stands "over" a specific dwelling. The corpus contains 40 occurrences, every one of them zodiacal or geometric; none locates a celestial body above a specific terrestrial point, for which the invariant astronomical choice is ὑπὲρ ("above" a zodiacal band or horizon). Feature D (ἔρχομαι): the Star "arrives". The corpus contains 356 occurrences with a celestial subject; every destination governed by ἐπί, εἰς or πρός is another body or a point in the chart, never a terrestrial geographic point.

Each feature was searched as a set of exact wordforms covering the full paradigm of the lemma, one form at a time, with diacritic-sensitive matching and full results paging: 164 forms for ἵστημι and 171 for ἔρχομαι across all four suppletive stems,

including the aorist stem that supplies Matthew's own ἐλθών. Each hit was read in full context and classified by two independent criteria: (1) syntactic, is the grammatical subject a celestial body?; (2) semantic, does the usage match the specific sense in Matthew 2:9? All classification decisions are recorded passage-by-passage in Supplementary Table 8, enabling independent verification by any reader with TLG access.

Statistical framework and Bayes factors

Fisher's exact test (one-sided) was applied to each 2×2 contingency table of corpus-hit classifications. The Laplace-smoothed probability under H_{astro} for each feature is $p_i = 1/(n_i + 1)$, where n_i is the number of corpus hits and the numerator 1 reflects zero positive matches. The joint Fisher combined statistic $X = -2\sum_i \ln p_i = 33.76$ on $\text{df} = 8$ gives $p_{\text{Fisher}} \approx 4.5 \times 10^{-5}$. Feature independence was checked by Gaussian copula simulation ($2{\times}10^6$ Monte Carlo draws) and Brown's variance-corrected method across equicorrelation values $\rho = 0$–0.95; the joint p remains below 0.05 across the whole tested range, to $\rho = 0.95$ (Supplementary Table 11).

The guiding-star comparison corpus comprises four independently attested pre-Matthean or contemporaneous traditions: Virgil *Aeneid* 2.687–711; the Timoleon flame tradition (Plutarch *Tim.* 8 and Diodorus Siculus 16.66.3, counted as one tradition from a common source); Lucian *Navigium* 7, 9; and the Israelite pillar-of-fire tradition (Exodus 13:21–22; Numbers 9:15–23). The *Protoevangelium of James* 21 is excluded as a derivative of Matthew 2:9. An omen corpus (Josephus *Jewish War* 6.289; Dio Cassius 54.29.8; PGM I.54–95) is included in sensitivity analyses. Each tradition was scored on features A–D by narrative function; scores are given in Supplementary Table 12. The corpus Bayes factor is:

$$\text{BF}_{\text{corpus}} = \prod_{i \in \{A,B,C,D\}} \frac{P(\text{feature}_i \mid H_{\text{guide}})}{P(\text{feature}_i \mid H_{\text{astro}})}, \tag{1}$$

where each probability uses Laplace smoothing over the respective corpus ($N = 4$ guiding-star traditions; $N = 24$ astronomical texts), giving $\text{BF}_{\text{corpus}} \approx 3.0 \times 10^6$ under strict independence, falling to $\approx 6 \times 10^1$ once both dependence corrections are applied at their strongest (Supplementary Table 14).

An independent within-Matthew Bayes factor was derived from Matthew's own Gospel. Every occurrence of the four features outside Matthew 2:9 was identified from the BibleHub interlinear (Strong's numbers G4254, G2476, G1883, G2064), cross-checked against NA28/SBLGNT lemmatization. Across $n = 5 + 20 + 7 + 86 = 118$ other Matthean occurrences, zero use any of the four words in an astronomical sense. The Laplace-smoothed likelihood ratio for feature i under H_{guide} versus H_{astro} is $\text{BF}_i = n_i + 1$, giving $\text{BF}_{\text{Matthew}} = 6 \times 21 \times 8 \times 87 = 87{,}696$. Because this analysis draws entirely from Matthew's Gospel while $\text{BF}_{\text{corpus}}$ draws from the TLG and guiding-star traditions, the datasets are disjoint and the factors are treated as conditionally independent, as they derive from non-overlapping corpora and distinct sampling frames. Their product is the joint Bayesian update from the linguistic evidence considered

here. Because both factors are products over the same four features, and those features are four properties of a single clause, the result is reported as a sensitivity range rather than a point estimate: $\mathrm{BF}_{\mathrm{joint}}$ spans 6.4×10^3 to 2.6×10^{11} across the dependence assumptions tabulated in Supplementary Table 14, with a central range of 2.5×10^5 to 6.0×10^7. Every value tested exceeds the Jeffreys threshold for decisive evidence by at least an order of magnitude. Features A–D co-occur in the same verse and are treated as conditionally independent in both BF calculations. Under positive correlation (plausible, since all four features reflect the same register choice), features absent together in the corpus tend to remain absent together, so $P(\text{data} \mid H_{\mathrm{astro}})$ increases relative to the product of marginals, raising the BF denominator and thereby *reducing* $\mathrm{BF}_{\mathrm{corpus}}$. The independence assumption therefore gives an upper bound on $\mathrm{BF}_{\mathrm{corpus}}$; the Gaussian copula simulation confirms the evidence remains substantial up to inter-feature correlation $\rho = 0.95$.

Software and reproducibility

All orbital-mechanics and statistical analyses were performed in Python 3.12 using `astropy`[20] (coordinate transformations, ephemeris access, time systems), `NumPy`[21] (vectorised array operations), `SciPy`[23] (Fisher's exact test, Newton–Raphson utilities), and `matplotlib`[24] (figures). Code and data needed to reproduce all figures and tables are available at https://github.com/adairaar/sob_astro_test. Key scripts include: `sob_proof_fine.py` (529-million-configuration heliocentric survey); `sob_mc_refined.py` (Monte Carlo refinement); `sob_epoch_robustness.py` (multi-epoch analysis); `corpus_pvalue.py` (Fisher combined p-value and Gaussian copula independence robustness test); `guiding_star_bayes.py` (corpus and within-Matthew Bayes factor computation). Sky-position results at representative epochs were independently cross-validated against Stellarium planetarium software[25], which implements the same Keplerian propagation model; computed positions agreed to within observational tolerances at all tested epochs.

Use of artificial intelligence

An AI assistant (Claude, Anthropic) was used to write and revise the analysis code, to suggest statistical treatments including the Kish design-effect correction and the Gaussian copula sensitivity analysis, and to assist in drafting and editing text. All code was reviewed and executed by the author, all reported quantities were checked against independent calculation, and all interpretive claims and final wording are the author's own. The code is available as described under Code availability, so that every number reported here can be reproduced independently.

Results

Matthew's own vocabulary disfavors the astronomical reading

A necessary condition for the astronomical reading of Matthew 2:9 is that the four motion words can carry astronomical senses. The author's own usage elsewhere in his Gospel provides direct evidence against this: across all 118 other occurrences of

προάγω, ἵστημι, ἐπάνω, and ἔρχομαι in Matthew, zero appear in any sense compatible with an astronomical body (Methods; full passage data in Supplementary Note 26 and Supplementary Table 16).

Προάγω appears five more times (14:22; 21:9×2; 26:32; 28:7), always transitive with a human direct object, identical syntax to προῆγεν αὐτούς in Matt 2:9; the intransitive positional-antecedence sense the astronomical reading requires is unattested in Matthew, nor does Matthew use other terms for "leading" that have an astronomical register (e.g., the intransitive προηγέομαι). Ἵστημι appears 20 more times with persons or objects at named physical locations; the mathematical stationary point is never its referent anywhere in the Gospel. Ἐπάνω appears seven more times denoting physical contact or immediate proximity; at Matt 28:2 an angel sits ἐπάνω the rolled-away stone, a heavenly being in point-contact with a terrestrial object, the same register as Matt 2:9. Ὑπέρ would be the expected Greek preposition for an astronomical body. On its own, read literally, ἐπάνω disqualifies the astronomical reading: a celestial body cannot be "over" a specific house in the sense of physical contact or immediate proximity. Nonetheless, I will conservatively only consider word counts in Matthew's own idiolect, not the semantic content of the words themselves, in the Bayes factor calculation below. Ἔρχομαι appears 86 more times with persons or agents arriving at terrestrial locations; never with a celestial body reaching a zodiacal position or celestial coordinate. Another verb could be used for "arriving" in astronomical texts (e.g., καταντάω), but Matthew never uses it.

Using Laplace smoothing, the Bayes factor for feature i from Matthew's own idiolect is $\mathrm{BF}_i = n_i + 1$ (non-astronomical uses n_i, zero astronomical uses). The joint within-Matthew Bayes factor is:

$$\mathrm{BF}_{\mathrm{Matthew}} = \prod_i (n_i + 1) = 6 \times 21 \times 8 \times 87 = 87{,}696. \tag{2}$$

Matthew's own vocabulary disfavors the astronomical reading at ≈88,000 to 1.

The key epistemic point is what $\mathrm{BF}_{\mathrm{Matthew}}$ measures: not that Matthew never discusses astronomy elsewhere, he rarely has occasion to, but that when describing the Star's motion he consistently reaches for words the external corpus identifies as belonging to the non-astronomical register. Because Matthew rarely describes the sky, this internal test is the weaker of the two linguistic tests: it shows only that the four words keep their ordinary senses everywhere else in his Gospel, so an astronomical sense would have to be unique to 2:9. The external corpus test supplies what the internal one cannot, the usage of writers who did describe the sky. This reading is not a modern analytical artifact: no ancient or medieval commentator interpreted the Star's motion in Matt 2:9 as a natural astronomical phenomenon[26]; every pre-modern interpreter understood the Star as not astronomical or astrological, but miraculous. The following sections test the remaining escape route: whether colloquial or imprecise language could nonetheless describe a real celestial event.

Geometric and temporal constraints from Matthew 2:9

The Magi traveled from Jerusalem (Old City center: 31°46′ N, 35°14′ E) to Bethlehem (Church of the Nativity: 31°42′ N, 35°12′ E), approximately 8 km along the ancient ridge road to the south-southwest. The straight-line bearing is 203°; because the road deviates around the central ridge, the actual azimuth of travel sweeps from approximately 190° to 210° along the route. I therefore adopt a road-corridor guidance criterion: the Star's azimuth must remain within $[190°, 210°] \pm 5°$ throughout the 2-hour journey (Fig. 1).

The criterion for προηγεν ("was going before") is not merely positional: a stationary object at the right azimuth cannot "go before" (let alone guide) travelers[9]. I therefore require the Star to exhibit perceptible southward motion throughout the guidance phase. For motion measured against the stellar background, as in the grid survey, I adopt an angular-velocity floor of $\omega_{\mathrm{thresh}} = 2° \, \mathrm{h}^{-1}$ (4× the Moon's motion relative to background stars, $\approx 0.5° \, \mathrm{h}^{-1}$, which Babylonian astronomers monitored routinely). This threshold is deliberately generous: naked-eye resolution is approximately 1′, so motion is detectable over a 30-minute observation at any rate above $\approx 0.03° \, \mathrm{h}^{-1}$, thirty-fold below this floor. Choosing any stricter threshold only strengthens the null result. The *guidance criterion* is therefore: (i) azimuth within the road corridor for the full journey, and (ii) apparent angular velocity $> 2° \, \mathrm{h}^{-1}$ throughout.

Walking pace on hilly terrain yields a journey time of approximately 2 hours. For the grid survey I locate this journey on a date in 5 BCE (Julian Day 1 719 755.9, ≈08:00–10:00 local solar time), the window proposed for the most recent candidate[8]; the close-flyby campaigns search the night instead (Table 1). Following arrival at Bethlehem, the Star must appear stationary over the specific house (ἐστάθη ἐπάνω, "stood over"). The verb ἐστάθη is passive aorist, denoting a punctual *change of state*: the Star transitioned into standing still, not that it was stationary from the outset. An object whose angular rate falls below ω_{thresh} throughout the entire observation window, including during the guidance phase, cannot satisfy this aorist stopping event, because no transition from motion to rest has occurred. The stopping criterion therefore carries an implicit sequential requirement: the angular rate must have been *above* ω_{thresh} during the guidance phase and must fall *below* it afterward. Guidance before stopping, in that order, is a formal constraint of the analysis. The *stopping criterion* is maximum apparent angular velocity below $2° \, \mathrm{h}^{-1}$ during the 10:00–12:00 window immediately following the journey, the same threshold ω_{thresh}, applied as an upper rather than lower bound. A sensitivity sweep across thresholds $\omega_{\mathrm{thresh}} \in \{0.25, 0.5, 1, 2, 5, 10\}° \, \mathrm{h}^{-1}$ yields zero satisfying configurations at every value tested (Supplementary Table 1).

Kinematic demonstration of incompatibility

The argument in this subsection is a mathematical demonstration, not a numerical result. The conceptual core is geometric: the south-southwest bearing of the Jerusalem–Bethlehem route means that any Keplerian body within the road corridor appears quasi-stationary throughout (satisfying the stopping criterion but not guidance), while any body with sufficient southward apparent motion lies outside the

corridor, a mutual exclusion. The analysis below makes this precise, establishing from the geometry of the bearing and Earth's rotation rate alone that the required ICRS angular velocities during guidance and stopping are unachievable for any heliocentric body, and that the transition between them is additionally forbidden by gravitational dynamics. The numerical surveys in the following subsections are corroborative.

Converting the two observational phases into ICRS velocity requirements via pre-computed East–North–Up (ENU) rotation matrices (Methods), the argument proceeds as follows.

*Stopping phase (*10:15–12:00*).* An object motionless in the local horizontal frame ("stood over") must, because the local AltAz frame co-rotates with Earth, sweep through ICRS at the sidereal rate. At a representative stopping altitude of 45° (az = 200°, Bethlehem latitude 31.7°N), this is $\omega_{\mathrm{stop}} = 14.76^\circ\,\mathrm{h}^{-1}$ (Table 2). Such a criterion already excludes fixed stars, novae, and planets, which all move at approximately the sidereal rate.

*Guidance phase (*08:00–10:00*).* An object visibly "going before" travelers at the minimum rate $2^\circ\,\mathrm{h}^{-1}$ in local AltAz along azimuth 200° yields, by vectorial addition of the sidereal rate, $\omega_{\mathrm{guide}} = 15.55^\circ\,\mathrm{h}^{-1}$ in ICRS.

Primary constraint (literal interpretation). Both ICRS values ($\approx 15^\circ\,\mathrm{h}^{-1}$) are unachievable for standard heliocentric bodies. Since $\omega_{\mathrm{ICRS}} = V_\perp/d$ with $V_\perp \propto d$, for any naked-eye heliocentric candidate with $d \gtrsim 0.1\,\mathrm{AU}$, the stopping state requires $V_\perp \gtrsim 1{,}070\,\mathrm{km\,s^{-1}}$ (e.g. $\approx 5{,}560\,\mathrm{km\,s^{-1}}$ at Mars opposition; $\approx 44{,}900\,\mathrm{km\,s^{-1}}$ at Jupiter), far above any known solar-system speed. Only in the very close flyby regime ($d \sim 0.003\,\mathrm{AU}$) does the requirement fall to $V_\perp \sim 30\,\mathrm{km\,s^{-1}}$, which is achievable in principle. Such an encounter can occur by day or by night, since a heliocentric orbit can approach Earth from any direction, and a body that close can move rapidly against the stars. The analytic argument does not exclude it. (For Matney's proposed comet specifically, the relevant window falls in full daylight.)

The close-flyby regime is thus the one case the analytic argument does not settle, and it is surveyed numerically in the section "Survey of close flybys". Full derivation and distance table in Supplementary Note 9.

Secondary constraint (astronomical-vocabulary interpretation). Even under the more permissive reading that the verbs describe motion relative to the stellar background, the transition between the two states is not available to a natural body. The guidance and sidereal-tracking trajectories differ in direction in ICRS, requiring an impulsive $\Delta v \approx 1.5\,\mathrm{km\,s^{-1}}$ at $d = 0.0026\,\mathrm{AU}$. Under Earth gravity ($a_\oplus = 2.6 \times 10^{-6}\,\mathrm{km\,s^{-2}}$) this requires $\approx 158\,\mathrm{h}$; the deceleration time scales as r^3, making the problem strictly worse at any greater distance. The corpus-linguistic analysis below independently closes this interpretive retreat by showing that the verbs belong to the narrative, not astronomical, register.

The kinematics also predicts what the numerical surveys should find, for reasons derived in Supplementary Notes 8, 9 and 29. A distant body keeps a nearly constant apparent pace over a few hours, by Kepler's second law, so anything fast enough to guide is equally fast afterward; it cannot stand still in the local sky without an unattainable transverse velocity; and a close flyby is carried westward by the sky, so that holding a bearing and then stopping requires its own motion to cancel the

Table 2 Kinematic parameters for the guidance and stopping phases at three representative stopping altitudes (azimuth 200°, Bethlehem). ω_{guide}: ICRS angular velocity of a star rising at the minimum detectable rate (2° h^{-1} in local AltAz) along azimuth 200°. ω_{stop}: ICRS angular velocity of a star stationary in local AltAz (sidereal tracking). Δv at $r = 0.0026$ AU (Matney's closest approach): minimum transverse velocity change required for the phase transition.

Alt. (°)	ω_{guide} (° h^{-1})	ω_{stop} (° h^{-1})	$\Delta\omega$ (° h^{-1})	$\omega_{\mathrm{stop}}/\omega_{\mathrm{thresh}}$	Δv (km s^{-1})
20	13.44	12.34	1.10	6.2×	2.07
45	15.55	14.76	0.79	7.4×	1.50
60	15.68	15.02	0.66	7.5×	1.24

diurnal drift twice, in order and at the same bearing. The surveys that follow test these predictions.

Exhaustive survey of heliocentric orbits

The null result is driven primarily by the geometric incompatibility between sustained southward motion and corridor confinement; the stopping criterion then eliminates the small residual near-matches that pass azimuth screening. I conducted an exhaustive heliocentric survey with a fine angular grid. The longitude of ascending node Ω and argument of perihelion ω were sampled at 5° intervals ($72 \times 72 = 5{,}184$ orientations per outer combination) across eccentricities $e \in \{0, 0.1, 0.2, 0.3, 0.5, 0.7, 0.8, 0.9, 0.95, 0.99, 1.0, 1.1, 1.5, 2.0, 5.0\}$ (elliptic through hyperbolic, spanning the eccentricity range of the known interstellar objects 1I/'Oumuamua ($e \approx 1.20$)[27] and 2I/Borisov ($e \approx 3.36$)[28]), perihelion distances $q \in \{0.02, 0.03, 0.04, 0.05, 0.07, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, 1.00\}$ AU (18 values, reaching Earth's orbital distance), inclinations 0° to 180° in 5° steps with additional fine sampling near 0° (42 values total; Supplementary Table 2), and perihelion epoch offsets $\Delta T \in \{-60, -40, -20, -10, -5, +5, +10, +20, +60\}$ days (9 values), yielding $18 \times 15 \times 42 \times 9 = 102{,}060$ outer combinations and $102{,}060 \times 5{,}184 = 529{,}079{,}040$ total configurations sampled. Of 202,372,076 ± 0.1% visible configurations (object above 5° altitude throughout the guidance window), zero simultaneously satisfy all three criteria: road-corridor azimuth, guidance motion $> 2°$ h^{-1}, and stopping motion $< 2°$ h^{-1} (Supplementary Table 20). Census counts of this kind carry a tolerance of order 0.1% because configurations lying within numerical noise of a threshold may be classed either way under different ephemeris implementations; the count satisfying all criteria is exactly zero and is reproduced identically across builds (Supplementary Note 22). The best azimuth corridor score is 0.015° inside the corridor (orbit $e = 1.1$, $q = 0.03$ AU, $i = 10°$, identified by the complementary fine-grid orbit inversion of the section "Orbit inversion confirms incompatibility"), yet that configuration's angular velocity is 0.23° h^{-1}, an order of magnitude below the required guidance motion. Configurations that come closest to the azimuth criterion are found to be nearly stationary: they satisfy the stopping criterion but have angular velocities far below

the $2^\circ\,\mathrm{h}^{-1}$ guidance-motion threshold, because a stationary object cannot προάγειν ("go before") anyone.

To verify this null result is not a grid-spacing artefact, I performed two Monte Carlo refinement sweeps (Supplementary Note 7 and Supplementary Figure 1). Sweep 1 (1.5 million configurations near the best-fit grid orbit) found zero of 601,150 visible configurations satisfying all criteria; the closest azimuth match had guidance velocity of $0.006^\circ\,\mathrm{h}^{-1}$, a factor of ∼330 below threshold. Sweep 2 (750,000 configurations near Matney's proposed 5 BCE comet) found zero of 294,672 visible configurations, with the best-azimuth candidate lying outside the road corridor entirely. No fine-tuning yields a viable solution: the incompatibility is structural.

Survey of Earth-orbiting objects

For geocentric objects the primary constraint of the section "Kinematic demonstration of incompatibility" does not apply: an Earth-orbiting body *can* in principle exhibit perceptible motion and then appear to slow, because its apparent motion is governed by its orbital velocity relative to the ground observer rather than by the sidereal rate. The geocentric case therefore requires a separate analysis in ground-frame AltAz coordinates.

The Moon, the only known natural geocentric body in antiquity, is no candidate in this frame: it moves with the diurnal rotation at up to $\approx 14.5^\circ\,\mathrm{h}^{-1}$ westward and never comes to rest in the local sky. Its motion against the stars (≈ 0.5–$0.6^\circ\,\mathrm{h}^{-1}$) is the relevant comparison for the star-frame grid survey, not here. A comprehensive survey of 28,800 hypothetical geocentric orbit configurations (the physically admissible subset (perigee altitude $> 100\,\mathrm{km}$) of 51,200 total grid configurations; semi-major axes from 7,000 km to 500,000 km, eccentricities 0–0.99, all inclinations and orientations; Supplementary Note 10), requiring guidance motion of 1–$5^\circ\,\mathrm{h}^{-1}$ within the road corridor at night followed by a stop within 4 h, finds zero configurations satisfying all criteria with a realistic stopping threshold $\omega_{\mathrm{stop}} < 0.20^\circ\,\mathrm{h}^{-1}$. The only configurations approaching even the loose threshold ($\omega_{\mathrm{stop}} < 0.50^\circ\,\mathrm{h}^{-1}$) are equatorial orbits near geosynchronous distance exhibiting the meridian-transit azimuth artifact: as any equatorial object crosses the meridian, its azimuth rate passes through zero briefly while altitude continues to change, a geometric effect rather than a physical deceleration. All 28,800 physically admissible geocentric configurations have the object above 5° altitude during at least one evaluated interval, so all contribute to the visible count. Combined with the heliocentric survey, 202,400,876 visible configurations spanning all orbit types yield zero solutions (Supplementary Table 20; Figs. 2 and 3).

It is worth being explicit about which requirement carries the result, because the criteria are not equally demanding when taken singly. Considered on its own, stopping is easy: in the fine survey 201,505,224 of the 202,372,076 visible configurations, better than 99.5%, are moving slowly enough at some moment to qualify. Nor is the difficulty simply that each criterion is individually rare. What no orbit achieves is the *conjunction*: guidance along the road corridor followed by a stop. Not one configuration in 5.29×10^8 manages both, and the failure is not marginal: the azimuth requirement alone admits zero configurations at every stopping threshold tested across

two orders of magnitude (Supplementary Table 1), and the best candidate located by direct optimization falls short of the required guidance interval by 0.04 h.

This matters for how the result should be read. A sceptical reader may prefer a looser construal of ἐστάθη (taking the Star merely to have lingered, rather than to have halted in the strict sense) and on such a reading the stopping criterion alone would be weak. The conjunction is untouched by that move. Whatever latitude is granted to "stood still," the object must first have guided travelers southward along a fixed bearing and then have ceased doing so, and it is the ordered pair of behaviors, not either member of it, that Keplerian motion cannot supply.

Figure 2 shows the two-criterion result: no configuration sustains the road-corridor direction *and* the guidance-motion requirement together. Figure 3 shows that adding the stopping criterion removes the single near-miss. The characteristic slope-1 diagonal in that figure is a rigorous consequence of Kepler's second law: apparent angular velocity $\omega_{\mathrm{app}} = V_{\perp}/d$ changes by at most $\lesssim 4\%$ over any four-hour window for bodies with $d \gtrsim 0.5$ AU (Supplementary Note 8), so the sole mechanism that could break this conservation, a retrograde stationary point, is geometrically excluded by the azimuth constraint.

Survey of close flybys

For a body passing within about 0.01 AU the analytic argument of the section "Kinematic demonstration of incompatibility" no longer applies, so close flybys were surveyed directly, in the ground frame and at night (Table 1). A nighttime survey of 10 million Monte Carlo orbits yields 66 genuine close flybys. None satisfies the guidance $\rightarrow$ stopping criterion (Supplementary Note 27). A numerical optimizer over 100 restarts reaches at most 0.58 h of contiguous guidance, against the 1.0-h requirement of that survey.

Because the guidance duration, the corridor and the meaning of "stopped" can each be read more loosely, a later campaign loosens all of them at once (Table 3). Of 10^9 sampled orbits, 6,116 make close flybys, and each is tested against 525 combinations: guidance of as little as 15 minutes, a corridor widened from 20° to 60°, an azimuth-drift tolerance raised from 1.5 to $30^\circ\,\mathrm{h}^{-1}$, and a stopping window of up to 8 hours. Stopping is judged by the angular displacement accumulated over an hour, as an observer comparing positions would judge it, rather than by an instantaneous rate, and the stop must occur at the bearing along which the object guided. No orbit passes at any combination; the tightest displacement achieved is 6.94° in an hour, against the loosest threshold of 5° (Supplementary Note 29).

Two further readings were tested. If the stop may occur at any bearing, one orbit of the 6,116 reaches 3.88°, but it is a hyperbolic body with an excess velocity of $52\,\mathrm{km\,s}^{-1}$, its total apparent motion never falls below $3.85^\circ\,\mathrm{h}^{-1}$, and it reaches that minimum 24° west of the corridor 3.8 hours after guidance ends. If the travelers left the road for the last stretch, the Star must turn with them. A turn east, toward the Church of the Nativity or anywhere from due south to due east, never occurs: all 2,024 orbits that pass through azimuths 120–210° move westward, so the eastern bearing always comes first. A turn west, from 210° to 270°, does occur, but no such orbit then stops; the tightest displacement is 9.22° in an hour.

Table 3 Loosened criteria in the close-flyby campaign (10^9 orbits, 6,116 close flybys), ground frame, at night. Each row lists how far a parameter was relaxed from the baseline; the result is for all combinations.

Parameter	Baseline	Loosest value tested
Guidance duration	2 h	15 min
Corridor width	20° (190–210°)	60° (150–210°)
Azimuth-drift tolerance in guidance	1.5° h^{-1}	30° h^{-1}
Stopping window after guidance	2 h	8 h
Stopping threshold (displacement per hour)	0.3°	5°
Final approach leg	none	0.25–0.5 h toward 90–180° or 210–270°, after a free turn of up to 1 h

Result: 0 of 525 single-leg combinations pass (tightest 6.94°); no eastward two-leg itinerary occurs; westward itineraries reach at best 9.22°. Flat priors give the same result (tightest 8.71°).

These are the outcomes the kinematics of the section "Kinematic demonstration of incompatibility" predicts: the cancellations needed to hold a bearing and then stop are brief and do not occur in the required order.

Orbit inversion confirms incompatibility

As a complementary test, I searched for the orbit that best fits the guidance-phase sky positions, then propagated it into the stopping window. Using a fine grid (5° spacing in Ω and ω) over the full eccentricity range ($e = 0$–5.0), the best-fitting guidance orbit achieves an azimuth corridor score of 0.015°, within the corridor, yet fails the guidance-motion criterion by an order of magnitude (0.23° h^{-1} versus the required 2° h^{-1}). When this orbit is propagated into the stopping window, the object rapidly sets below the horizon, the antithesis of the text's requirement (Supplementary Figure 5).

Corpus-linguistic analysis: Matthew 2:9 vocabulary is incompatible with Greek astronomical and astrological discourse

The kinematic result above is independently corroborated by a statistical analysis of ancient Greek astronomical vocabulary. If Matthew 2:9 described a real astronomical phenomenon, its language should conform to the technical register shared by Greek astronomical and astrological texts. I assembled a corpus of 24 texts (mathematical astronomy, astrological application, and didactic popularizations) comprising approximately 619,000 words, spanning the fourth century BCE through the fourth century CE (Supplementary Table 7; Methods).

I identified four genre-register features of Matthew 2:9 that any astronomical interpretation must implicitly assume, testing not merely lexical choice but whether the narrative events they encode appear anywhere in astronomical writing:

(A) προάγω (*proagō*): a star leading human travelers toward a destination.

(B) ἵστημι (*histēmi*): a celestial body coming to rest *above a terrestrial place*, as distinct from the zodiacal station the verb denotes in technical usage.
(C) ἐπάνω (*epanō*): rather than ὑπὲρ: a celestial body localized above a specific terrestrial address.
(D) ἔρχομαι (*erchomai*): a celestial body "arriving" at a specific location on Earth.

All four Matthew-type usages score zero in the 619,000-word corpus (Table 4; passage-level citations in Supplementary Table 8). Feature A: all 11 corpus occurrences of προάγω carry positional or temporal senses ("preceding in longitude"), never narrative guidance ($p = 0.083$). Feature B: all 121 occurrences of ἵστημι place a celestial body in a zodiacal sign, a degree, an aspect, a chart angle or beside a fixed star, and never above a house, city or person ($p = 0.008$); Matthew's own form ἐστάθη does not occur in the corpus at all. Feature C: all 40 occurrences of ἐπάνω carry geometric or zodiacal senses, never localizing a body above a named place ($p = 0.024$). Feature D: ἔρχομαι is used constantly of celestial bodies arriving, 356 occurrences, but what they arrive at is always another body or a point in the chart ("Mars *coming upon* the Sun"; "Jupiter *coming into* the horoskopos"), never a place on the ground ($p = 0.003$; a thematic robustness check yields the same result at every feature, Supplementary Table 9). The joint Fisher combined statistic is $\chi^2(8) = 33.8$, $p \approx 4.5 \times 10^{-5}$ (Fig. 4); Gaussian copula correction (2×10^6 Monte Carlo draws) maintains significance across the whole range of inter-feature correlation tested, to $\rho = 0.95$, and every drop-one-feature subset remains jointly significant ($p < 0.05$).

Table 4 Corpus linguistics: Fisher's exact test of Matthew 2:9 vocabulary against Greek astronomical and astrological texts.

Feature	Matthew usage	Corpus count	Matthew	p (Fisher)
(A) προάγω	Guidance of travelers	0 / 11	1	0.083
(B) ἵστημι	Celestial body halts above a place	0 / 121	1	0.008
(C) ἐπάνω	Above specific earthly point	0 / 40	1	0.024
(D) ἔρχομαι	Spatial arrival of Star	0 / 356	1	0.003
Joint (A–D)	—	—	—	$\chi^2(8) = 33.8$, $p \approx 4.5 \times 10^{-5}$

Corpus: 24 texts, ≈619,000 words, 4th c. BCE–4th c. CE (see Supplementary Table 7). TLG counts, 2026-09-04 (MIT IRIS proxy), by exact wordform search over the full paradigm of each lemma, with Exact Match and Diacritics sensitive enabled; see Supplementary Table 8 for passage-level citations. Fisher's exact test, one-sided: $p = 1/(N + 1)$, where N = corpus negatives. "Corpus count" shows Matthew-type positive / non-Matthew-type negative hits. All four Matthew-type usages score zero in the corpus. Ἔρχομαι is suppletive; all four stems (ἐρχ-, ἐλευσ-, ἐλθ-/ἠλθ-, ἐληλυθ-) were searched, including the aorist that supplies Matthew's own ἐλθών. Joint row reports Fisher's combined $\chi^2 = -2\sum \ln p_i = 33.8$ with df = 8, $p \approx 4.5 \times 10^{-5}$; corrected for inter-feature correlation by Gaussian copula: $\rho = 0.5$ yields $p = 0.0027$; significance maintained across the whole tested range, to $\rho = 0.95$ (Supplementary Note 19 and Supplementary Table 11).

The null result is uniform across genre: even astrological texts (Ptolemy's *Tetrabiblos*, Vettius Valens, Hephaestion), the most likely to adopt narrative vocabulary,

use φέρεσθαι/κινεῖσθαι and ὑπὲρ throughout; Matthew's guidance vocabulary is absent from mathematical and astrological texts alike (Supplementary Note 17 and Supplementary Table 8). This result is independently corroborated by Schironi's philological survey of ancient Greek astronomical terminology[29], which identifies as the standard technical vocabulary for celestial leading/preceding: ἡγούμενος/προηγούμενος (not προάγω); and for a body located above a zodiacal band: ὑπέρ (not ἐπάνω). For stationary points Schironi gives στηρίζω/στηριγμοί, and the corpus shows ἵστημι serving the same technical sense: Geminus glosses planets "standing still" as what "they call στηριγμούς". The distinction that bears on Matthew 2:9 is accordingly not which verb is chosen but what it governs, and in the corpus that is never a terrestrial place. The documentary horoscopes and astronomical papyri from Oxyrhynchus[30] confirm the same conventions in applied astrological practice, extending the vocabulary evidence beyond literary texts to working astronomical documents.

Bayes factor: Matthew 2:9 vocabulary matches the guiding-star narrative tradition

The corpus analysis above establishes that Matthew's four-feature vocabulary is improbable under the astronomical hypothesis (Fisher combined $p \approx 4.5 \times 10^{-5}$). A complementary question is whether that vocabulary is probable under the alternative hypothesis that Matthew 2:9 belongs to the ancient guiding-star narrative tradition (H_{guide}). Four independent traditions attest this genre: Virgil's *Aeneid* 2.687–711 (Servius *ad Aen.* 2.694 identifies the guide as Venus/Lucifer, the morning star); Plutarch and Diodorus Siculus on the Timoleon guiding light; Lucian's *Navigium* 7, 9 (explicitly a star); and the LXX pillar-of-fire (*Exodus* 13.21; *Wisdom* 10.17 describes it as star-like at night) [17, 26, 31]. Each involves a divinely-sent luminous guide. I scored each tradition for features A–D and estimated $P(\text{feature}_i \mid H_{\text{guide}})$ with Laplace smoothing ($k = 1$, $N = 4$; Supplementary Table 12). Feature A is present in all four traditions; D in three; B in two; C in two (Lucian's ἐπικαθίσαι τῷ καρχησίῳ and the Israelite pillar standing ἐπί the Tabernacle, both using the same ἐπι- root as Matthew's ἐπάνω). Under the independence approximation:

$$P(A, B, C, D \mid H_{\text{guide}}) = \prod_i P_i \approx 1.39 \times 10^{-1}. \tag{3}$$

The Bayes factor comparing the guiding-star hypothesis to the astronomical hypothesis is therefore:

$$\text{BF} = \frac{P(A, B, C, D \mid H_{\text{guide}})}{P(A, B, C, D \mid H_{\text{astro}})} = \frac{1.39 \times 10^{-1}}{4.67 \times 10^{-8}} \approx 3.0 \times 10^{6}. \tag{4}$$

Here $P(A, B, C, D \mid H_{\text{astro}}) = \prod_i p_i = 4.67 \times 10^{-8}$ is the product of the four individual Fisher exact probabilities, the correct joint data probability under independence, rather than the Fisher combined meta-test tail probability (4.5×10^{-5}), which is a test statistic, not a joint data probability. This BF exceeds 100 on the Jeffreys scale,

providing strong evidence in favor of the guiding-star narrative hypothesis over the astronomical one. Sensitivity analyses confirm robustness. Varying the corpus composition by adding four additional omen texts gives BF $= 5.1 \times 10^5$; varying the Laplace smoothing parameter $k \in [0.5, 3]$ gives BF $\in 2.3$–3.4×10^6; a $3\times$ denominator uncertainty yields a range of 1.0–9.3×10^6 (Supplementary Table 12).

Feature C (ἐπάνω) is particularly diagnostic: every astronomical and omen text examined uses ὑπὲρ for a celestial body above a terrestrial point, including Josephus (*JW* 6.289) and Dio Cassius (54.29.8), both of whom use ἵστημι/αἰωρέω with ὑπὲρ; ἐπι-based localization is attested in the present corpus only in legendary and sacred narrative texts (Lucian; *Joseph and Aseneth* 14.4; the Israelite pillar-of-fire tradition, where ἐπί marks the cloud standing over the Tabernacle: Exod 40.38, Num 9.15, 9.18–22). This distributional split confirms the Bayes factor at the individual-feature level (Supplementary Note 24).

Discussion

The foregoing analysis produces three convergent results. *First*, Matthew's own vocabulary establishes an internal prima facie case: across 118 other Matthean occurrences of the four motion words, zero appear in an astronomical sense, yielding $\mathrm{BF}_{\mathrm{Matthew}} \approx 88{,}000$ against the astronomical reading from the author's idiolect alone. *Second*, orbital mechanics addresses the "loose-language" escape route: no Keplerian body tested exhibits both the sustained guidance motion and the punctual stopping, a result supported by a kinematic argument, a survey of 529 million heliocentric and 28,800 geocentric configurations finding zero solutions, Monte Carlo refinement, and a targeted close-flyby search. *Third*, external corpus-linguistic analysis delivers the positive genre classification: Matthew's four-feature vocabulary is unattested in the surveyed ≈619,000-word astronomical corpus while fitting the narrative tradition at $\mathrm{BF}_{\mathrm{corpus}} \approx 3.0 \times 10^6$ under strict independence. Because the internal and external linguistic datasets are entirely disjoint, their Bayes factors are treated as conditionally independent of one another; both are nonetheless products over the same four features of a single clause, so the joint factor is reported as a sensitivity range. At the independence corner of Supplementary Table 14 it is $87{,}696 \times 3.0 \times 10^6 \approx 2.6 \times 10^{11}$; at the opposite corner, with within-feature clustering and between-feature dependence both at their strongest, it is $\approx 6.4 \times 10^3$. Every cell of the grid remains decisive, and the range is reported in full rather than collapsed to a single figure. This linguistic classification is independent of the orbital mechanics.

Previous studies operated within a *search* paradigm, identifying the best-matching historical event. The present work operates within a *demonstration* paradigm: deriving kinematic requirements from the text and testing the whole parameter space against them. A failed search leaves open the possibility of an undiscovered candidate; a survey of the whole space, here finding no satisfying configuration, addresses the entire class at once, within the criteria stated. The null result holds across reference dates (all major proposed epochs, 12 BCE to 1 BCE), guidance durations from 15 minutes to 2 hours, corridor widths from 20° to 60°, and stopping windows of up to 8 hours (Tables 1

and 3), and across every orbit type: elliptic, parabolic, hyperbolic, heliocentric, and geocentric.

A recent systematic review of proposals concluded that all purely astronomical candidates were effectively eliminated; this left a different sort of approach based on the Jupiter horoscope of 17 April 6 BCE (Molnar's astrological reading) as the "leading and default explanation"[10, 15]. The present analysis, however, extends the result beyond astronomical candidates to the astrological position as well, suggesting that the fault is not in the stars but in the astrological theories themselves. First, the kinematic demonstration applies equally to Jupiter during its 6 BCE apparition: the planet's apparent angular velocity ($\approx 0.035^\circ\,\mathrm{h}^{-1}$ relative to background stars, far below the $2^\circ\,\mathrm{h}^{-1}$ threshold) satisfies the stopping criterion throughout its entire synodic period: it cannot προάγειν ("go before") anything, let alone travelers, in any kinematically meaningful sense on the timescale the narrative implies. The same can be said for any other planet. Second, the astrological reading of Matt 2:9 depends on a "loose language" escape: it requires the Greek verbs to describe not the Star's apparent sky motion but an astrological configuration. The corpus-linguistic analysis closes this exit: προάγω, ἵστημι, ἐπάνω, and ἔρχομαι are absent from the astrological sub-corpus (Ptolemy *Tetrabiblos*, Vettius Valens, Hephaestion, Dorotheus, Pseudo-Manetho, Paul of Alexandria) in exactly the Matthew-type senses, confirming that the vocabulary belongs to the narrative guiding-star register rather than the astrological one. All of Molnar's deductions about the Greek of Matt 2:9, which Schaefer depends on, are without philological foundation[9, 17, 32]. Third, Schaefer characterizes the skeptical position, that Matthew 2:9 is legendary, as having "no evidence in its favor, and having explained none of the details or coincidences"[15]. This characterization overlooks a substantial body of New Testament scholarship: the majority critical position, represented by Brown[11] and reinforced in the same Brill volume by Merz, who co-authored the standard historical-Jesus reference work[13], holds that the infancy narratives are literary constructions rather than historical reports. The present paper provides that evidence quantitatively: a Bayes factor that is decisive on the Jeffreys scale under every dependence assumption tested, indicating a non-physical and a specifically legendary motif, and a kinematic analysis in which no surveyed object satisfies the text's requirements at any proposed epoch.

Non-gravitational forces cannot rescue the astronomical interpretation. 'Oumuamua's anomalous acceleration ($\approx 5 \times 10^{-9}\,\mathrm{km\,s^{-2}}$; [33]), the largest known, delivers at most $\Delta v \approx 3.6 \times 10^{-5}\,\mathrm{km\,s^{-1}}$ over 2 h, a factor of $\approx 4 \times 10^4$ below the $1.5\,\mathrm{km\,s^{-1}}$ required at closest approach ($d \approx 0.003$ AU), and $\approx 3 \times 10^7$ below the requirement at $d \gtrsim 0.1$ AU. No chemically plausible force bridges this gap in any orbital regime.

A common retreat reads the verbs loosely: προῆγεν as static "ahead of" and ἐστάθη as conveniently placed at journey's end. This is grammatically impermissible[9], and the corpus results undermine it regardless: the analysis tests whether the vocabulary belongs to the astronomical register, and it does not: προάγω, ἔρχομαι, ἵστημι with a terrestrial complement, and ἐπάνω (rather than ὑπὲρ) are all anomalous in astronomical writing. A looser reading reinforces, not dissolves, the linguistic evidence.

The three lines of evidence are mutually reinforcing: the mechanics shows the described motion is not produced by any natural object tested; the internal linguistics shows Matthew never uses these words astronomically; the external linguistics shows the vocabulary belongs to a different register. Thematically, active stellar guidance to a specific address and stopping above it is unattested in the texts examined, while it recurs in the guiding-star legend tradition (Supplementary Information). Feature C (ἐπάνω) is particularly diagnostic: every astronomical and omen text in the present corpus employs ὑπὲρ for a celestial body above a terrestrial point; ἐπι-localization is attested in the present corpus only in legendary narrative texts (Supplementary Information).

The result bears exclusively on Matthew 2:9. The companion verse Matthew 2:2 imposes additional constraints (a potential heliacal rising requires solar proximity) not analyzed here; my analysis is deliberately conservative in ignoring them. The conclusion rests on the assumption shared by every naturalistic proposal in the literature: that the narrative intends a physically real object whose motion can be described in astronomical terms. Within that assumption, the present analysis finds no viable astronomical candidate across any eccentricity class or orbit type.

Five boundaries deserve emphasis. First, the orbital result is conditional on the reading of Matthew 2:9 that it tests: a star that goes before the travelers along their road and then stands over a particular place. Other readings have been proposed, in which the Star did not single out a dwelling, or in which προάγω means only that it lay ahead of them, and the orbital analysis does not constrain those. Two considerations bear on how far they weaken the result. The criteria were loosened well beyond the literal reading (Table 3), and the linguistic evidence indicates that the literal reading is the text's own: the four verbs carry their ordinary narrative senses in Matthew and are absent in those senses from the astronomical corpus, and every premodern interpreter read the Star's motion as miraculous[26]. Allegorical or symbolic readings are likewise not constrained. Second, the parameter space, though large, is finite, and the margins are stated rather than assumed. On the displacement criterion the tightest in-corridor case is 6.94° in an hour against a loosest threshold of 5°, a margin of 1.39×; a reader prepared to call a drift of about 7° an hour, some fourteen lunar diameters, a standstill would find candidates, although it is difficult to see how an observer would describe such motion as stopping. Third, non-Keplerian physics is considered but not exhaustively modeled: the primary constraint (transverse velocities $\sim 10^3$–$10^4\,\mathrm{km\,s^{-1}}$) is insensitive to any chemically plausible non-gravitational force, but arbitrarily exotic objects are not formally excluded. Fourth, Matthew 2:2 is excluded from the orbital survey; that verse may independently constrain candidate objects or further indicate other culturally relevant details. Fifth, nothing can be determined about the origins of the narrative, if it was inherited by the Gospel author or was their own invention, from this analysis. Independent studies are needed to address source-critical questions, including the adoption of a prior "morning star" tradition concerning Jesus[31].

The demonstration framework introduced here is generalizable beyond Matthew 2:9. The companion verse Matthew 2:2 independently constrains any candidate to solar-proximity orbits (a heliacal rising requires the object to have been near

the sun in preceding weeks), a restriction deliberately excluded from the present survey whose addition would only strengthen the null result. More broadly, any ancient text whose astronomical interpretation requires specific celestial kinematics can in principle be subjected to the same two-step analysis: an exhaustive orbital parameter-space survey establishing what is kinematically possible, followed by a corpus-register test establishing whether the vocabulary belongs to the astronomical genre. Ancient omen literature, East Asian annals used to date historical comets, and other sky-event narratives for which both physical constraints and comparative textual corpora exist are natural applications of this combined framework.

Data availability

The Greek corpus search results, TLG hit records, and Monte Carlo output data supporting this study are available in the GitHub repository at https://github.com/adairaar/sob_astro_test.

Code availability

All Python scripts used for the orbital-mechanics surveys, Monte Carlo analyses, and corpus-linguistic calculations are available at https://github.com/adairaar/sob_astro_test. Full reproduction instructions are provided in the repository `README`.

Author contributions

A.A. conceived the study, designed and implemented all orbital-mechanics and corpus-linguistic analyses, produced all figures and tables, and wrote the manuscript.

Competing interests

The author declares no competing financial or non-financial interests.

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Figure legends

Fig. 1 Observational constraints from Matthew 2:9. a) Polar sky chart showing the required apparent motion: during the 2-hour guidance phase (προῆγεν, "was going before"), the Star must move visibly southward with azimuth within the road corridor $[190°, 210°]$ toward Bethlehem; it then appears stationary over a specific house (star symbol, ἐστάθη, "stood over"). Altitude is not constrained during stopping (any altitude above the horizon is acceptable): scholarly interpretations of "stood over" range from just above the horizon to zenith. b) The road corridor $[190°, 210°]$ in the local sky from Bethlehem. The straight-line bearing is 203°; the road deviates around the central ridge, sweeping between 190° and 210°.

Fig. 2 Direction and guidance motion: the two criteria are not met together. Each point is one visible orbital configuration from the full heliocentric parameter space (broad survey) and two Monte Carlo refinement sweeps. The horizontal axis (log scale) shows the maximum azimuth deviation of the object from the road corridor $[190°, 210°]$ during the guidance phase (0 = perfectly in corridor); the vertical axis (log scale) shows the minimum angular velocity during that same phase (must exceed $2° \, h^{-1}$ to προάγειν). A configuration satisfying both direction *and* guidance motion simultaneously would fall in the gold-shaded upper-left region (azimuth deviation $< 5°$ *and* motion $> 2° \, h^{-1}$). That region is essentially empty: green points (corridor orbits, azimuth deviation $< 5°$) cluster along the bottom-left, quasi-stationary throughout with guidance speeds $\sim 0.006° \, h^{-1}$ (factor $\sim$330 below threshold); fast-moving orbits (upper right) are far outside the road corridor. The single annotated outlier ($e = 2.0$, $\Delta\text{az} = 2.1°$, motion $= 2.89° \, h^{-1}$) does enter the gold region but fails the third (stopping) criterion, as shown explicitly in Fig. 3.

Fig. 3 No surveyed orbit satisfies Matthew 2:9. Each point represents one visible orbital configuration sampled from the full heliocentric parameter space (broad survey) and two Monte Carlo refinement sweeps; non-corridor points are coloured by eccentricity (plasma scale). The horizontal axis (log scale) shows the *minimum* angular velocity during the guidance phase (08:00–10:00 local time; must exceed $2° \, h^{-1}$ to lead the Magi southward). The vertical axis (log scale) shows the *maximum* angular velocity during the subsequent stopping phase (10:15–12:00; must fall *below* $2° \, h^{-1}$ to appear stationary over Bethlehem). A configuration satisfying προάγειν and ἐστάθη simultaneously would occupy the gold-shaded *lower-right* region; that region contains zero orbits across every eccentricity class. The purple diagonal band is the *Keplerian constraint*: the 1st–99th percentile of the empirical stopping-to-guidance speed ratio, spanning only $\times 0.995$–$\times 1.033$ around $y = x$. Over any two-hour window the apparent angular velocity $\omega_{\text{app}} = V_{\perp}/d$ changes by at most $\lesssim 4\%$ for heliocentric bodies (Supplementary Note 8); all points therefore cluster invisibly close to the diagonal. The required region lies orders of magnitude from this band, making the exclusion structural rather than a consequence of sparse sampling. Green points (azimuth deviation $< 5°$, in-corridor) cluster near the origin: quasi-stationary throughout both phases, confirming the anti-correlation visible in Fig. 2. The single annotated outlier ($e = 2.0$, $\Delta\text{az} = 2.1°$) reaches $2.89° \, h^{-1}$ during guidance but retains the same speed during stopping, placing it on the diagonal and well above the stopping threshold. Supplementary Figure 1 shows each MC sweep separately.

Fig. 4 Linguistic incompatibility of Matthew 2:9 with Greek astronomical discourse. TLG counts, 2026-09-04, by exact wordform search over the full paradigm of each lemma. **Left**, Corpus hit counts for each lemma (blue bars, 11–356 occurrences); the Matthew-type usage (red square) scores zero in every case: no hit in the 24-text corpus ($\approx$619,000 words) belongs to the narrative sense Matthew employs. **Right**, Fisher's exact test: $-\log_{10}(p)$ for each of the four features; dashed orange line marks $p = 0.05$, dotted red line the Bonferroni threshold ($\alpha/4 = 0.0125$). Features B (ἵστημι), C (ἐπάνω) and D (ἔρχομαι) individually cross $p = 0.05$, and B and D also cross the Bonferroni threshold; the joint Fisher combined statistic is $\chi^2(8) = 33.8$, $p \approx 4.5 \times 10^{-5}$, and every drop-one subset remains jointly significant. The corpus includes both mathematical astronomy and astrological texts; the absence of Matthew's vocabulary is uniform across all genres.